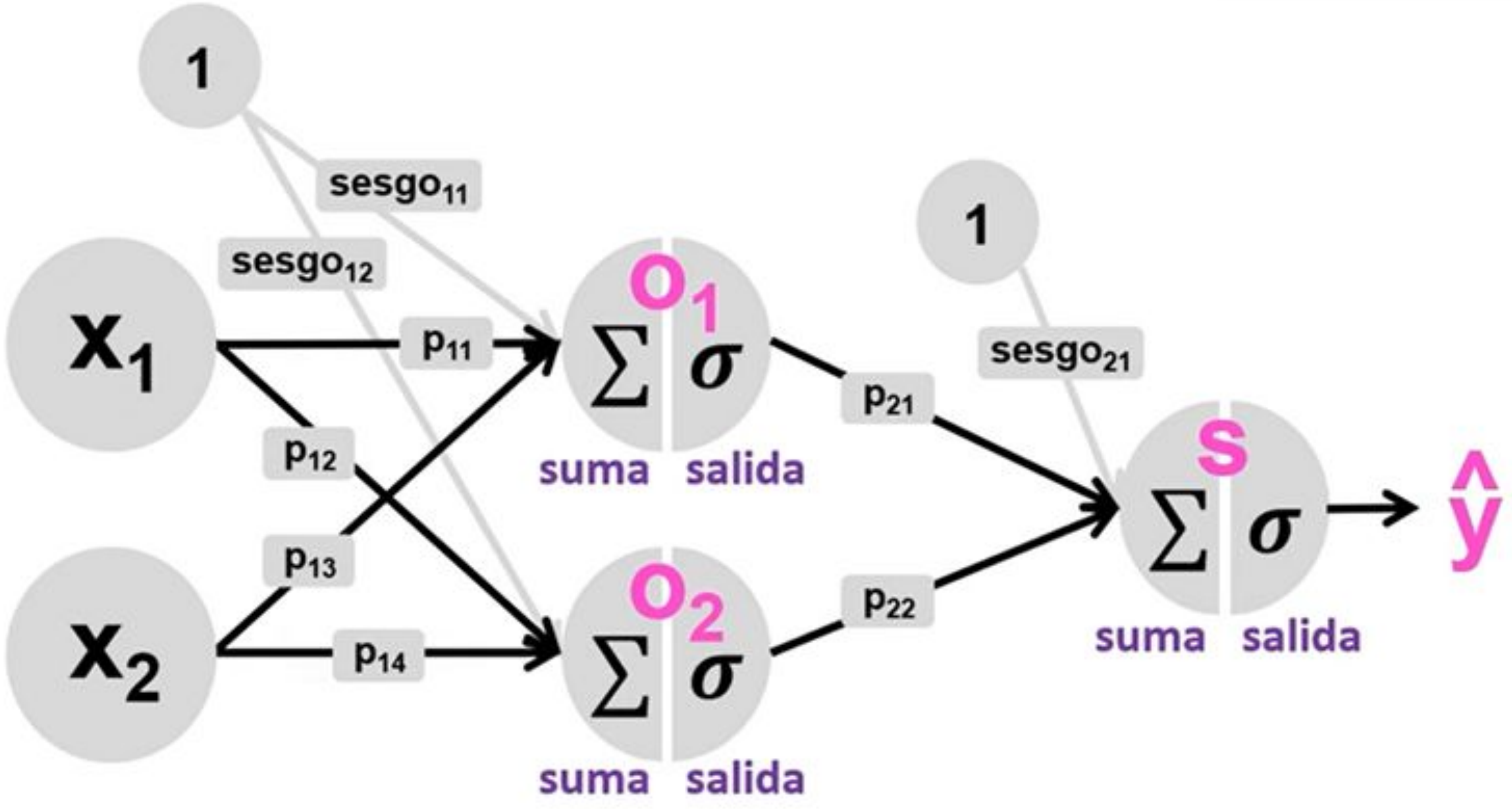
Backpropagation y Gestión de modelos

01 Backpropagation

Algoritmo de Retropropagación (Backpropagation)

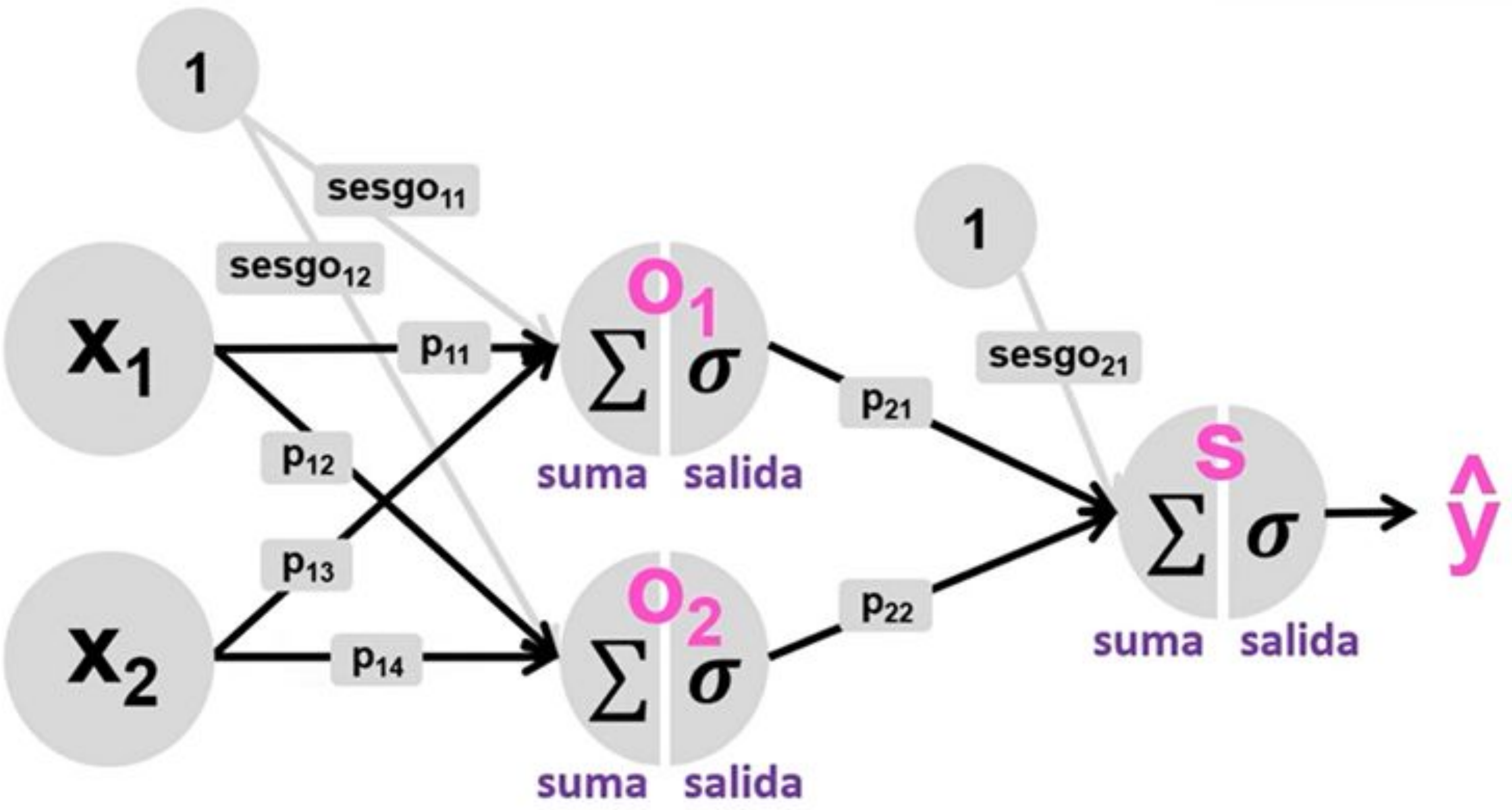
El algoritmo de retropropagación (backpropagation) es uno de los métodos más importantes para entrenar redes neuronales artificiales. Fue popularizado en la década de 1980 y se utiliza para **minimizar la función de error (o función de pérdida) ajustando los pesos de la red.**



Funcionamiento

Cálculo del gradiente para el peso$_{21}$

$$\frac{\partial E}{\partial p_{21}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial p_{21}}$$



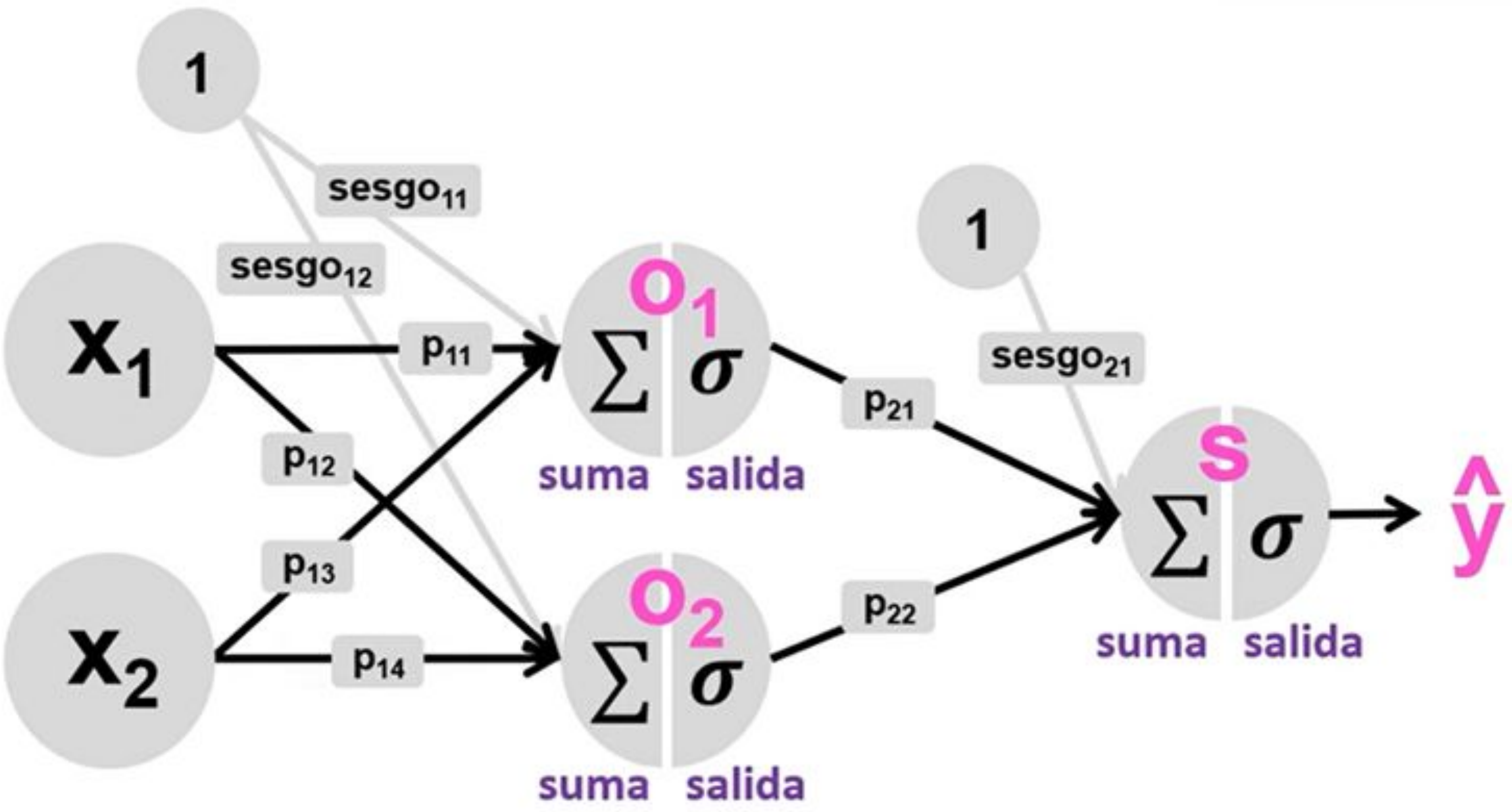
Funcionamiento

Cálculo del gradiente para el **peso$_{21}$**

$$\frac{\partial E}{\partial p_{21}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial p_{21}}$$

$$\frac{\partial E}{\partial \hat{y}} = \hat{y} - y$$

La función de error es el error cuadrático.



Funcionamiento

Cálculo del gradiente para el peso$_{21}$

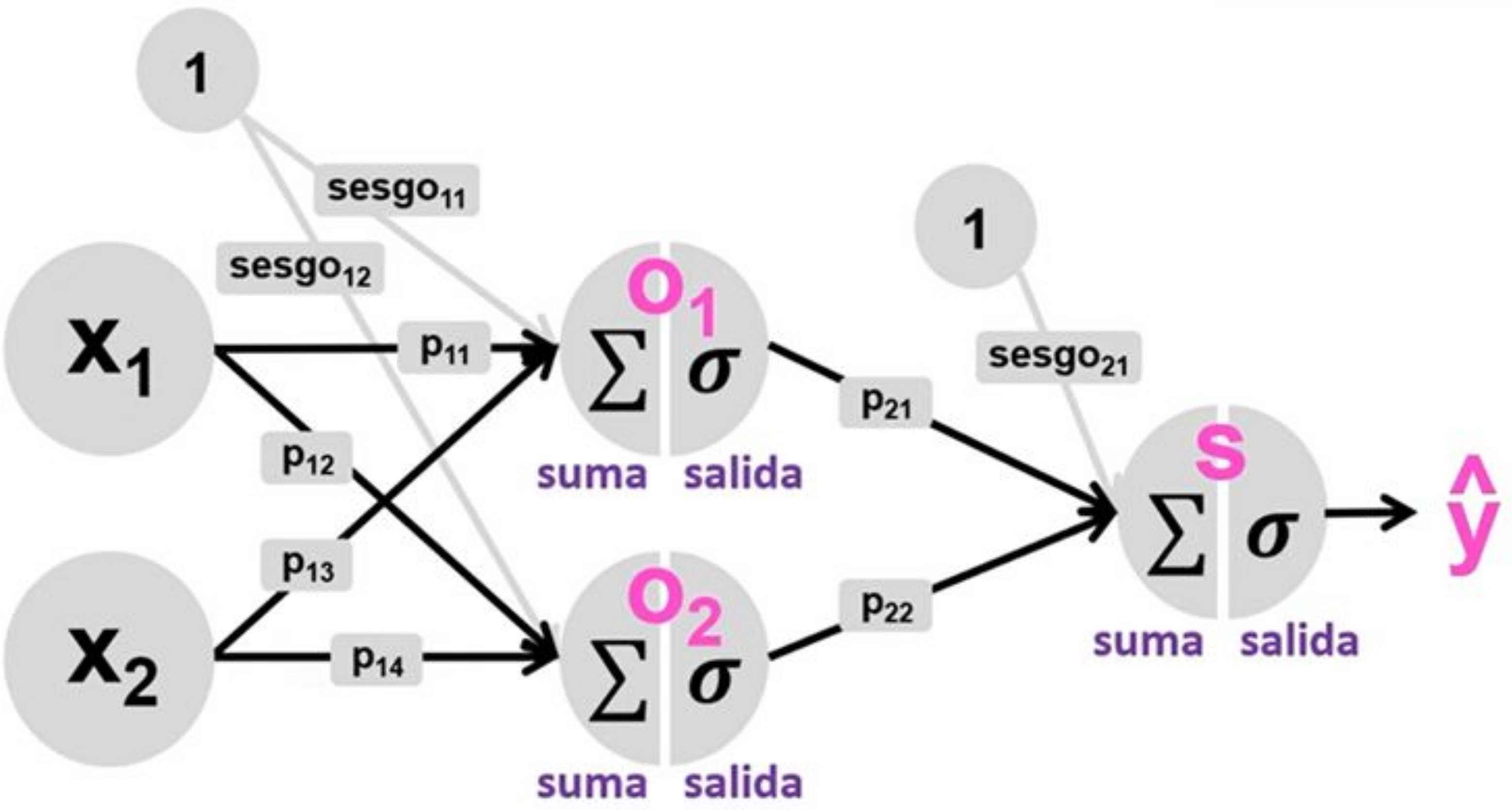
$$\frac{\partial E}{\partial p_{21}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial p_{21}}$$

$$\frac{\partial E}{\partial \hat{y}} = \hat{y} - y$$

La función de error es el error cuadrático.

$$\frac{\partial \hat{y}}{\partial suma_s} = \hat{y}(1 - \hat{y})$$

La función de salida es la función sigmoide.



Funcionamiento

Cálculo del gradiente para el peso$_{21}$

$$\frac{\partial E}{\partial p_{21}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial p_{21}}$$

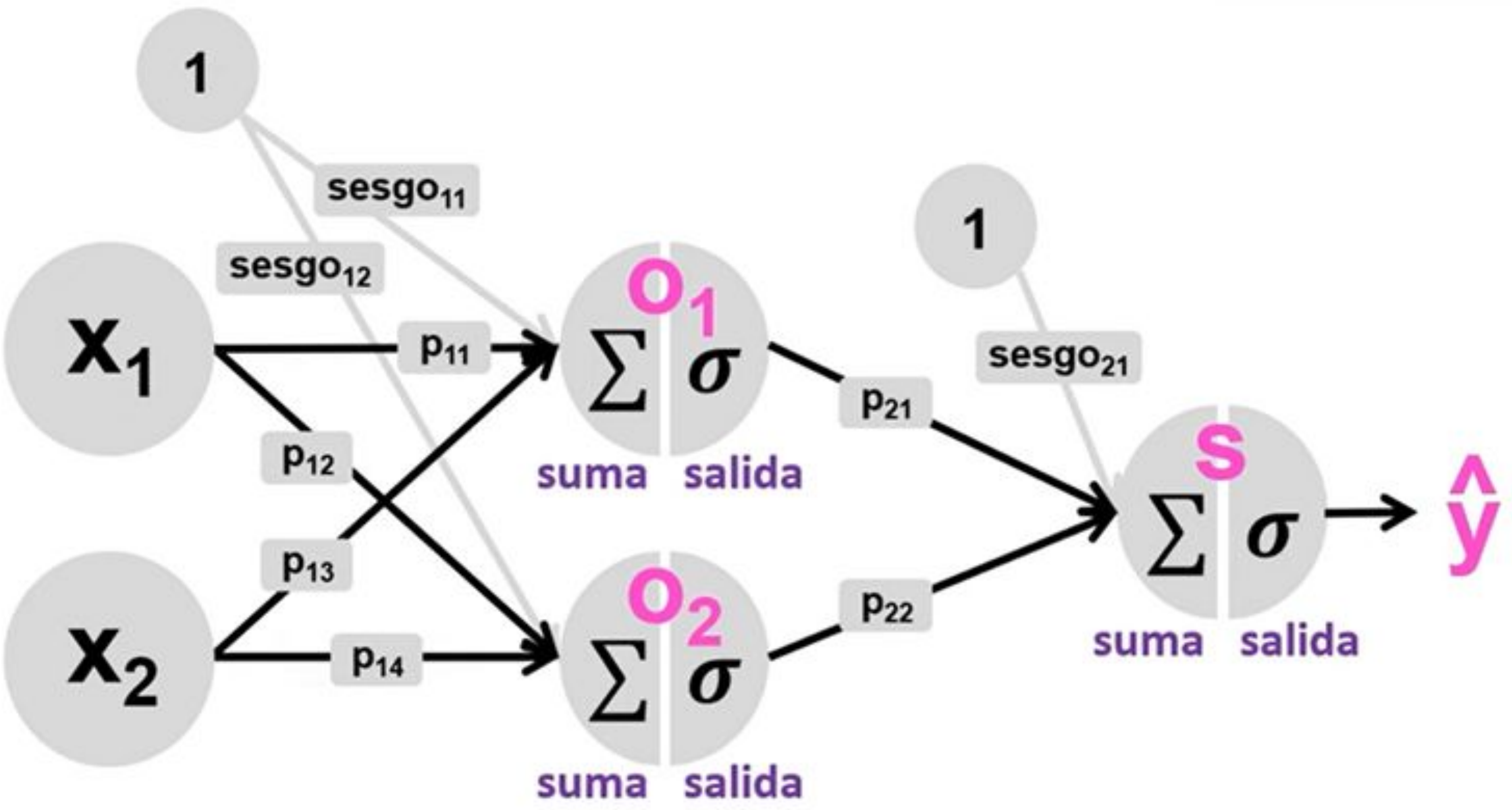
$$\frac{\partial E}{\partial \hat{y}} = \hat{y} - y$$

La función de error es el error cuadrático.

$$\frac{\partial \hat{y}}{\partial suma_s} = \hat{y}(1 - \hat{y})$$

La función de salida es la función sigmoide.

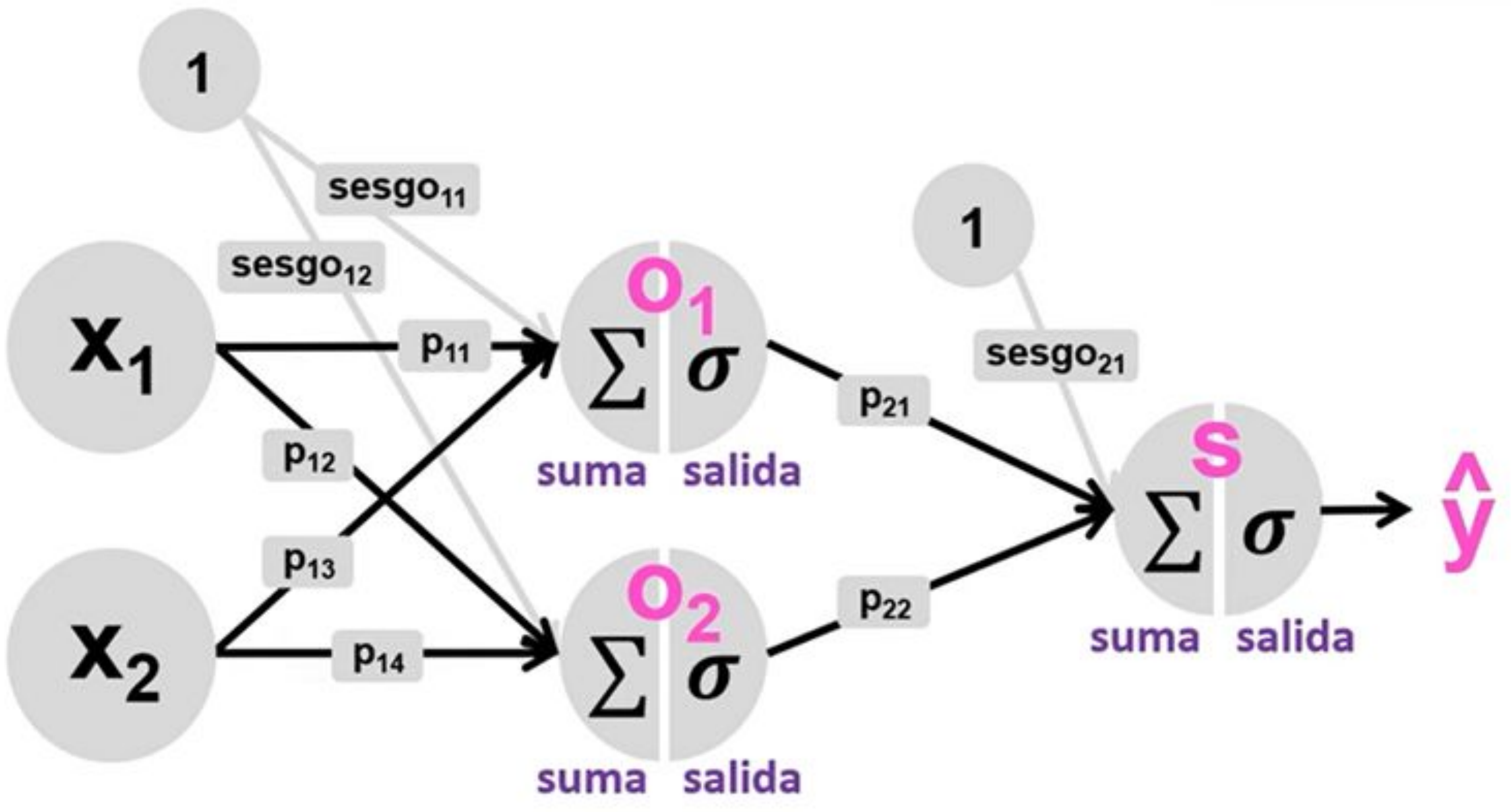
$$\frac{\partial suma_s}{\partial p_{21}} = \frac{\partial\, salida_{01} p_{21} + salida_{02} p_{22} + sesgo_{21}}{\partial p_{21}} = salida_{01}$$



Funcionamiento

Cálculo del gradiente para el peso$_{21}$

$$\frac{\partial E}{\partial p_{21}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial p_{21}}$$

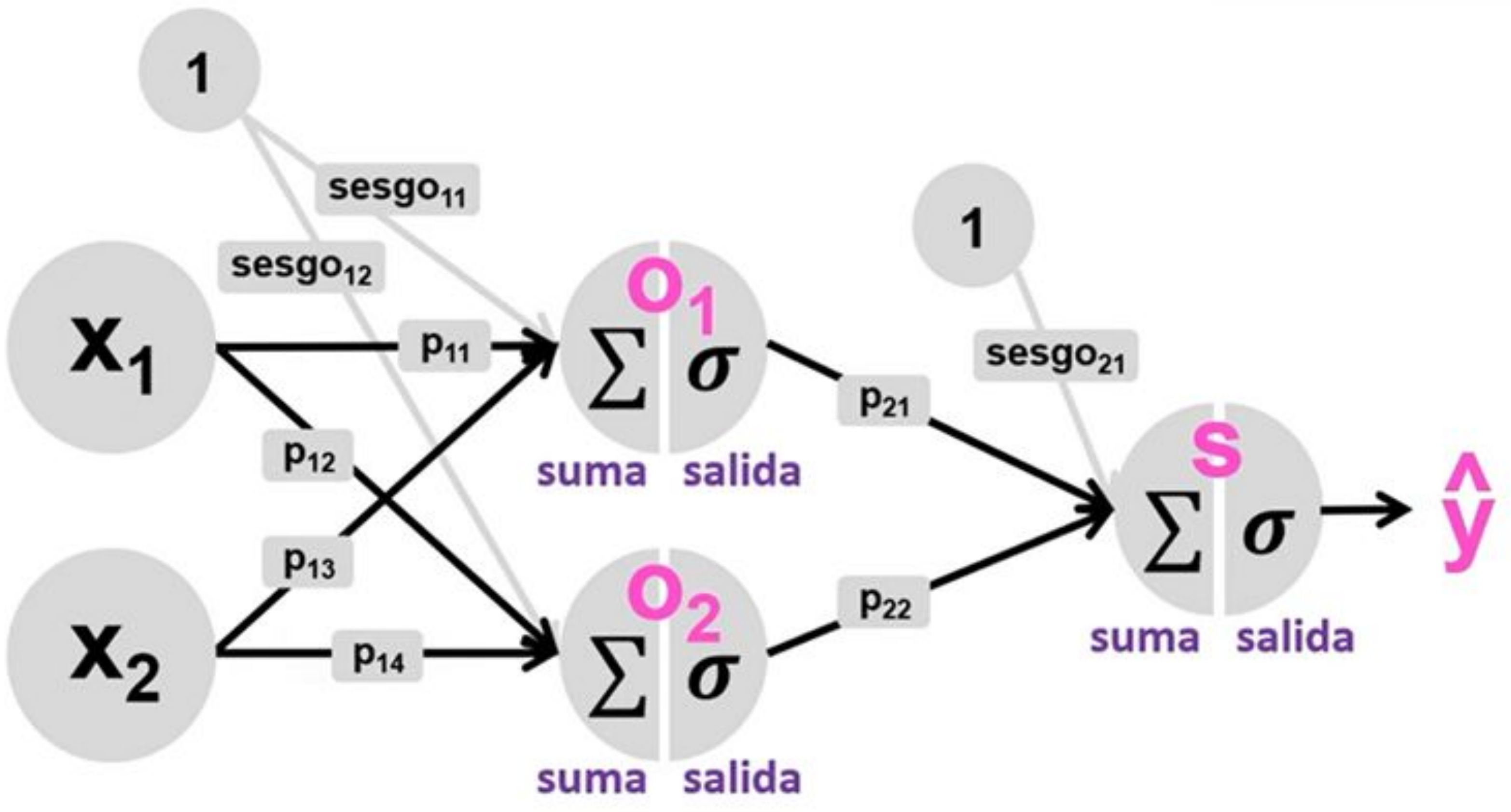


$$\frac{\partial E}{\partial p_{21}} = \underbrace{(\hat{y} - y) . \hat{y}(1 - \hat{y})}_{\delta_s} . salida_{01} = \delta_s \, salida_{01}$$

Funcionamiento

Cálculo del gradiente para el peso$_{21}$

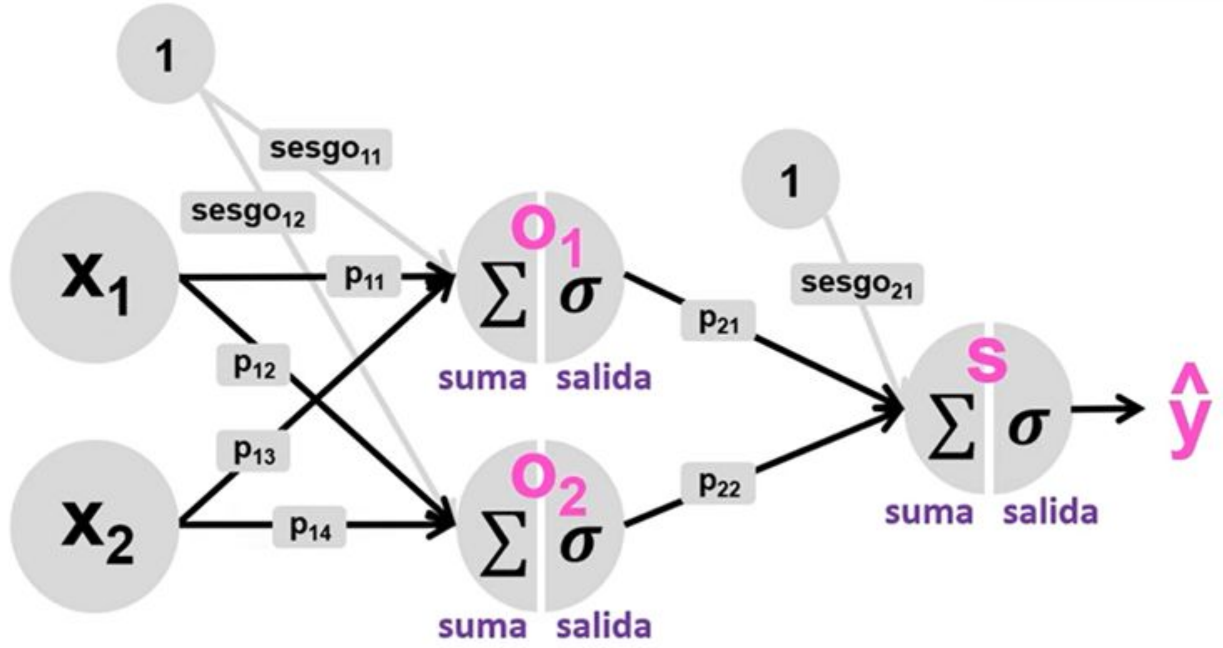
$$\frac{\partial E}{\partial p_{21}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial p_{21}}$$



$$\frac{\partial E}{\partial p_{21}} = \underbrace{(\hat{y}-y).\ \hat{y}(1-\hat{y})}_{\delta_s}\ .\ salida_{01} = \delta_s\ salida_{01}$$

$$\frac{\partial E}{\partial p_{22}} = \delta_s\ salida_{02} \qquad \bigg| \qquad \frac{\partial E}{\partial sesgo_{21}} = \delta_s 1$$

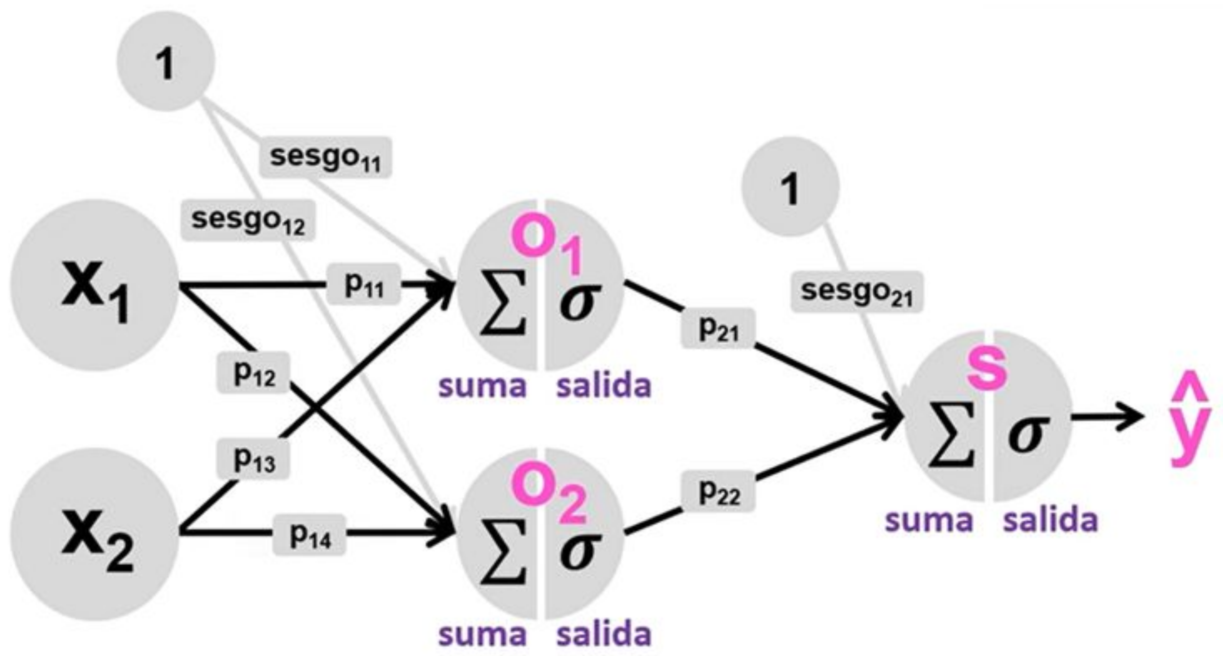
Funcionamiento



Cálculo del gradiente para el peso$_{11}$

$$\frac{\partial E}{\partial p_{11}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial salida_{01}} * \frac{\partial salida_{01}}{\partial suma_{01}} * \frac{\partial suma_{01}}{\partial p_{11}}$$

Funcionamiento

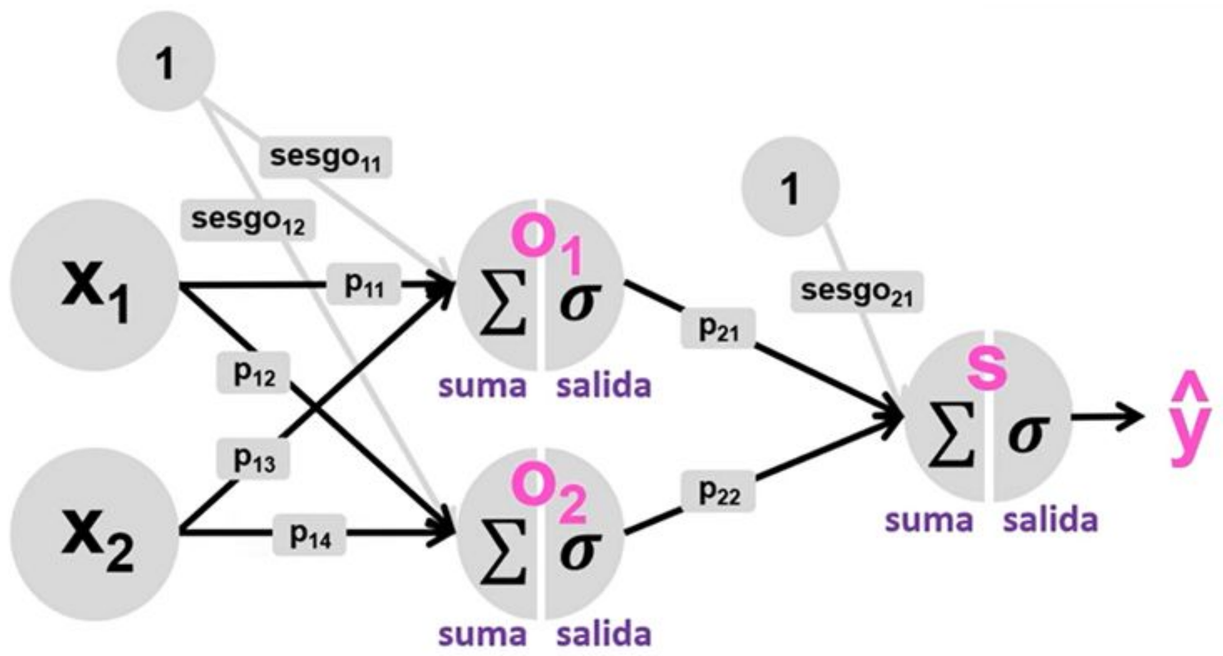


Cálculo del gradiente para el peso$_{11}$

$$\frac{\partial E}{\partial p_{11}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \boxed{\frac{\partial suma_s}{\partial salida_{01}}} * \frac{\partial salida_{01}}{\partial suma_{01}} * \frac{\partial suma_{01}}{\partial p_{11}}$$

$$\boxed{\frac{\partial suma_s}{\partial salida_{01}}} = \frac{\partial \, salida_{01} p_{21} + salida_{02} p_{22} + sesgo_{21}}{\partial salida_{01}} = p_{21}$$

Funcionamiento



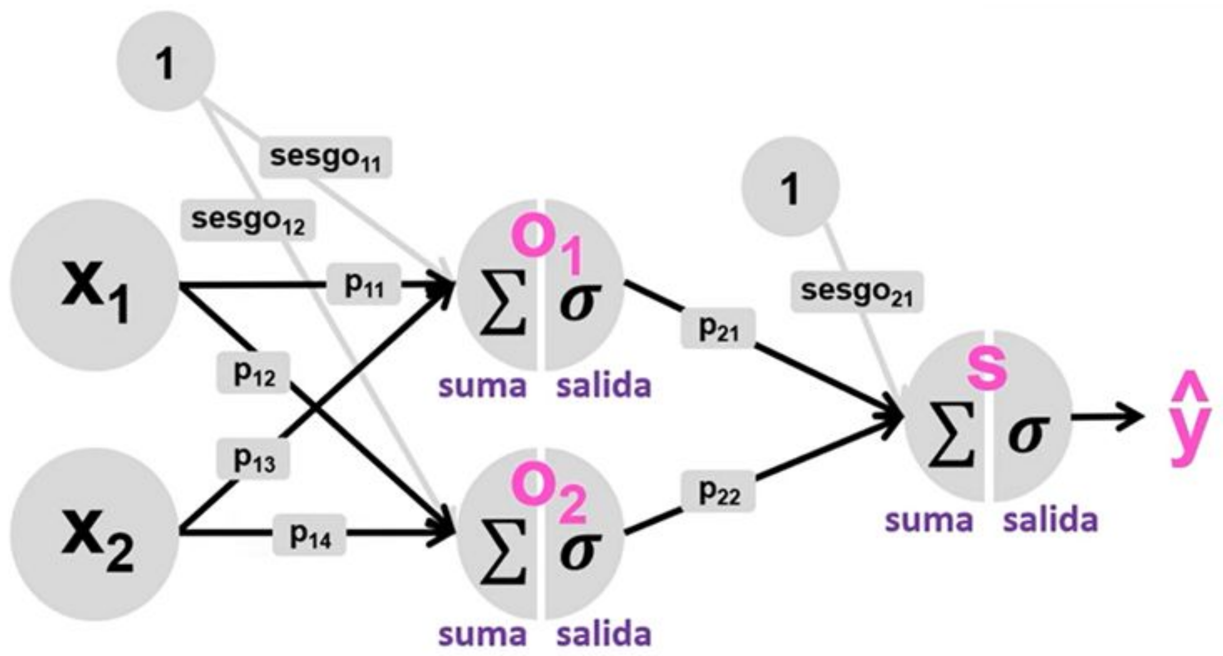
Cálculo del gradiente para el peso$_{11}$

$$\frac{\partial E}{\partial p_{11}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial salida_{01}} * \frac{\partial salida_{01}}{\partial suma_{01}} * \frac{\partial suma_{01}}{\partial p_{11}}$$

$$\frac{\partial suma_s}{\partial salida_{01}} = \frac{\partial\, salida_{01} p_{21} + salida_{02} p_{22} + sesgo_{21}}{\partial salida_{01}} = p_{21}$$

$$\frac{\partial salida_{01}}{\partial suma_{01}} = salida_{01}(1 - salida_{01})$$

Funcionamiento



Cálculo del gradiente para el peso$_{11}$

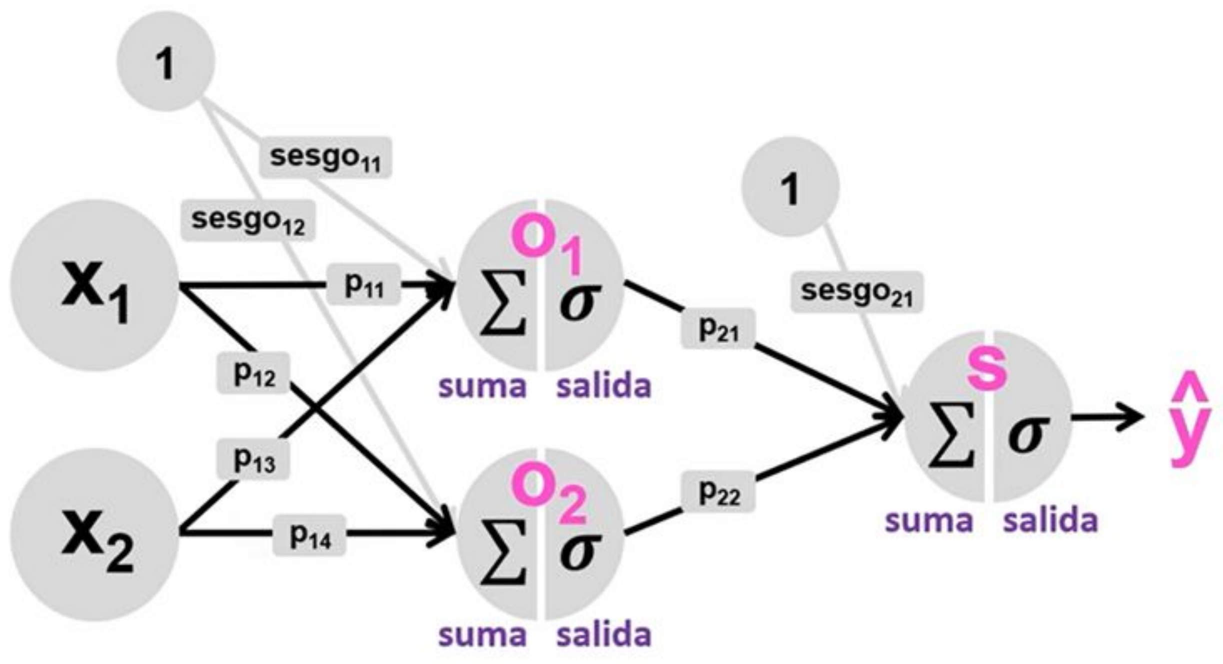
$$\frac{\partial E}{\partial p_{11}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial salida_{01}} * \frac{\partial salida_{01}}{\partial suma_{01}} * \frac{\partial suma_{01}}{\partial p_{11}}$$

$$\frac{\partial suma_s}{\partial salida_{01}} = \frac{\partial\, salida_{01} p_{21} + salida_{02} p_{22} + sesgo_{21}}{\partial salida_{01}} = p_{21}$$

$$\frac{\partial salida_{01}}{\partial suma_{01}} = salida_{01}(1 - salida_{01})$$

$$\frac{\partial suma_{01}}{\partial p_{01}} = \frac{\partial\, x_1 p_{11} + x_2 p_{13} + sesgo_{11}}{\partial p_{11}} = x_1$$

Funcionamiento

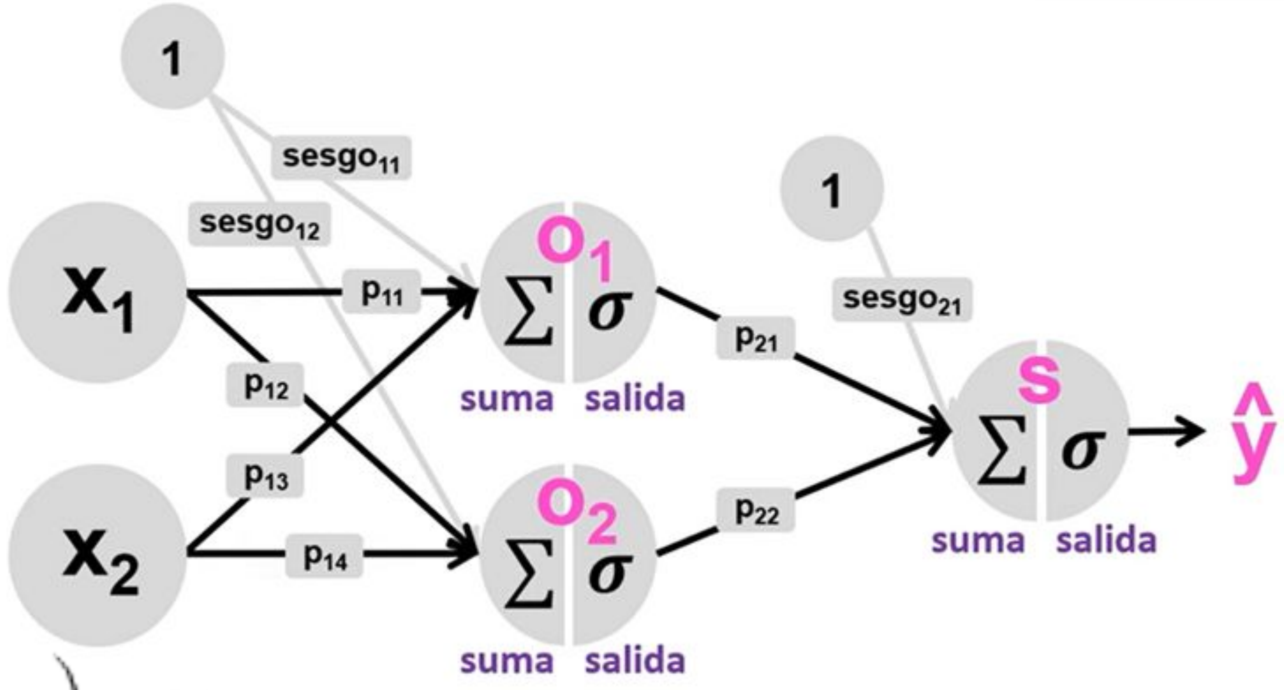


Cálculo del gradiente para el peso$_{11}$

$$\frac{\partial E}{\partial p_{11}} = \frac{\partial E}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial suma_s} * \frac{\partial suma_s}{\partial salida_{01}} * \frac{\partial salida_{01}}{\partial suma_{01}} * \frac{\partial suma_{01}}{\partial p_{11}}$$

$$\frac{\partial E}{\partial p_{11}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{21}.\, salida_{01}(1 - salida_{01}).\, x_1$$

Funcionamiento



Obtenemos los gradientes para los demás pesos y sesgos.

$$\frac{\partial E}{\partial p_{11}} = (\hat{y}-y).\,\hat{y}(1-\hat{y}).\,p_{21}.\,salida_{01}(1-salida_{01}).\,x_1$$

$$\frac{\partial E}{\partial p_{13}} = (\hat{y}-y).\,\hat{y}(1-\hat{y}).\,p_{21}.\,salida_{01}(1-salida_{01}).\,x_2$$

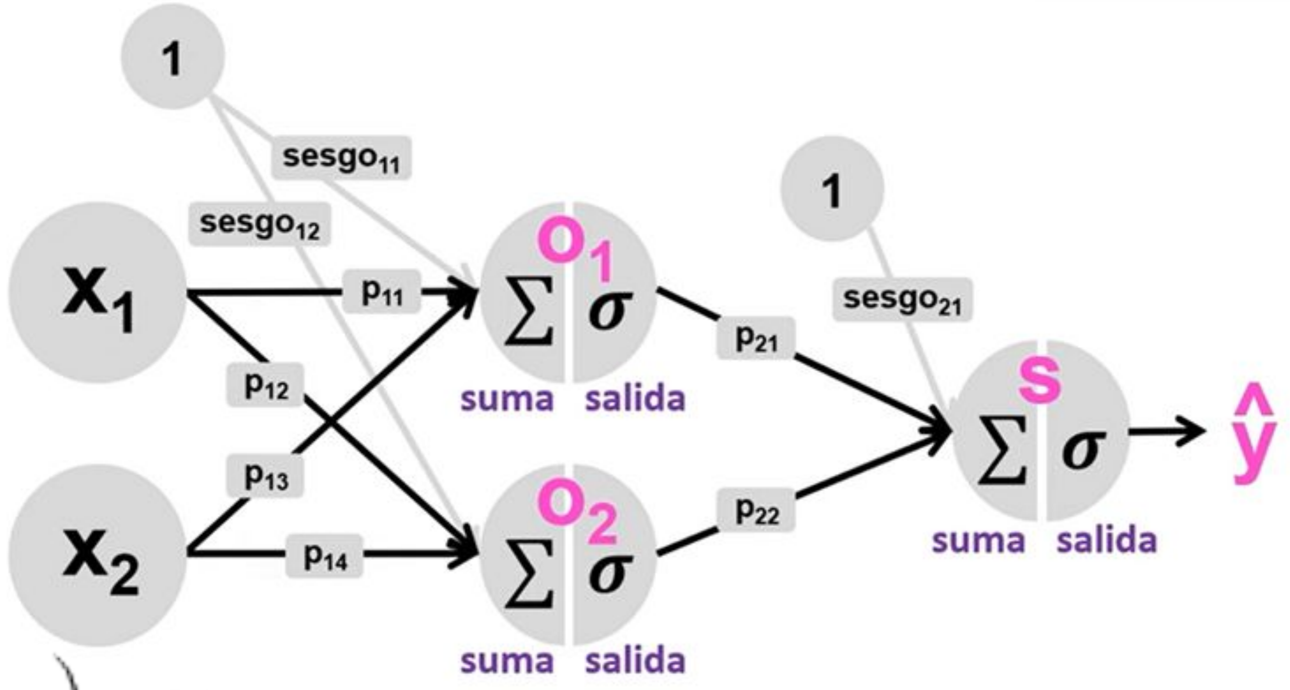
$$\frac{\partial E}{\partial p_{12}} = (\hat{y}-y).\,\hat{y}(1-\hat{y}).\,p_{22}.\,salida_{02}(1-salida_{02}).\,x_1$$

$$\frac{\partial E}{\partial p_{14}} = (\hat{y}-y).\,\hat{y}(1-\hat{y}).\,p_{22}.\,salida_{02}(1-salida_{02}).\,x_2$$

$$\frac{\partial E}{\partial sesgo_{11}} = (\hat{y}-y).\,\hat{y}(1-\hat{y}).\,p_{21}.\,salida_{01}(1-salida_{01}).\,1$$

$$\frac{\partial E}{\partial sesgo_{12}} = (\hat{y}-y).\,\hat{y}(1-\hat{y}).\,p_{22}.\,salida_{02}(1-salida_{02}).\,1$$

Funcionamiento



Obtenemos los gradientes para los demás pesos y sesgos.

$$\frac{\partial E}{\partial p_{11}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{21}.\, salida_{01}(1 - salida_{01}).\, x_1$$

$$\frac{\partial E}{\partial p_{13}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{21}.\, salida_{01}(1 - salida_{01}).\, x_2$$

$$\frac{\partial E}{\partial p_{12}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{22}.\, salida_{02}(1 - salida_{02}).\, x_1$$

$$\frac{\partial E}{\partial p_{14}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{22}.\, salida_{02}(1 - salida_{02}).\, x_2$$

$$\frac{\partial E}{\partial sesgo_{11}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{21}.\, salida_{01}(1 - salida_{01}).\, 1$$

$$\frac{\partial E}{\partial sesgo_{12}} = (\hat{y} - y).\, \hat{y}(1 - \hat{y}).\, p_{22}.\, salida_{02}(1 - salida_{02}).\, 1$$

$$\boxed{p_{11}^{nuevo} = p_{11}^{viejo} - \eta \frac{\partial E}{\partial p_{11}}}$$

$$\boxed{sesgo_{11}^{nuevo} = sesgo_{11}^{viejo} - \eta \frac{\partial E}{\partial sesgo_{11}}}$$

¡Muchas gracias!

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